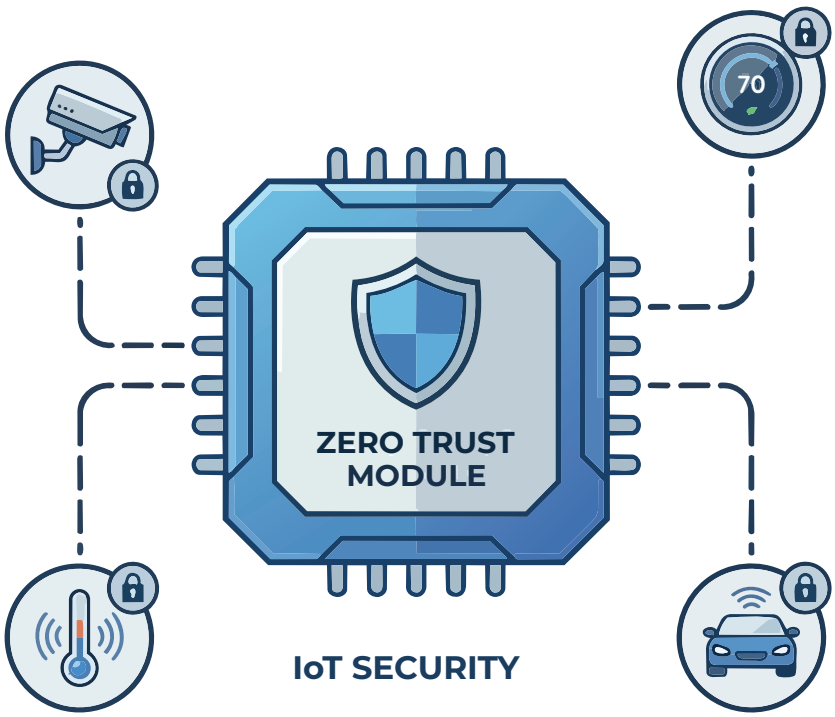
Zero Trust Module for IoT

It's critical to build Zero Trust into IoT networks and replace outdated trust models. ns-3 simulations help safeguard vulnerable smart devices.



Traditional network security functioned like a fortress with a castle-and-moat perimeter. Once a device crossed the perimeter firewall or connected through a gateway, it enjoyed implicit trust across the internal network. This conventional security model falls flat in modern Internet of Things (IoT) deployments. Smart sensors, actuators, and connected appliances are routinely placed in physically accessible spots, run on minimal computing hardware, and transmit data across shared, open networks.

To overcome these vulnerabilities, the Zero Trust architecture establishes a straightforward rule: never trust, always verify. Under the NIST SP 800-207 guidelines, network location alone never grants privilege. Every access request, even from an internal sensor, is treated as if it originates from an untrusted or even compromised network.

The key tenets defined in NIST SP 800-207 applicable to IoT are:

- ***Resource protection:*** All data sources, sensor feeds, and computing services are treated as resources [NIST SP 800-207, Section 2.1, Tenet 1].
- ***Per-session trust evaluation:*** Trust is never permanent; access to individual resources is evaluated on a per-session and per-request basis with least-privilege permissions [NIST SP 800-207, Section 2.1, Tenet 3].
- ***Dynamic policy decisions***: Access is determined dynamically using identity, device status, and environmental conditions via a Policy Decision Point (PDP) and enforced at a Policy Enforcement Point (PEP) [NIST SP 800-207, Section 2.1, Tenet 4; Section 3].
- ***End-to-end encryption and authentication:*** All communications are secured, authenticated, and encrypted regardless of physical topology [NIST SP 800-207, Section 2.1, Tenet 2].

To facilitate reproducible Zero Trust experiments, the authors have developed and published the Zero Trust IoT library for ns-3 (zero-trust-iot), which is officially available on the NS-3 App Store. The library provides plug-and-play C++ components for cryptographic authentication, runtime context drift re-scoring, and in-line packet filtering. It allows network developers and academic researchers to simulate and benchmark real-world Zero Trust defences without investing in expensive physical sensor testbeds.

Usage of ns-3 in IoT applications

Network Simulator 3 (ns-3) is a widely adopted, discrete-event network